

Chilled Water Plant Spec Sheet

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Project: Lakefront Office Tower — Phase II

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SECTION 23 64 16 — CENTRIFUGAL WATER CHILLERS

1.0 SCOPE OF WORK

Furnish and install centrifugal water chillers, cooling towers, primary and secondary chilled water pumps, condenser water pumps, variable-flow primary-secondary piping system, and associated valves per drawings M3.01–M3.06. System designed for N+1 chiller redundancy; 1,400 tons total installed capacity.

2.0 EQUIPMENT SCHEDULE

Tag	Equipment	Capacity	Key Spec	Drive	Notes
CH-1	Centrifugal Chiller	700 TR	R-134a, IPLV 0.480 kW/Ton	VFD	Primary
CH-2	Centrifugal Chiller	700 TR	R-134a, IPLV 0.480 kW/Ton	VFD	Primary
CH-3	Centrifugal Chiller	700 TR	R-134a, IPLV 0.480 kW/Ton	VFD	N+1 Standby
CT-1	Induced Draft Cooling Tower	875 TR	CW 85/95°F, 7°F approach	100 HP fan	Lead
CT-2	Induced Draft Cooling Tower	875 TR	CW 85/95°F, 7°F approach	100 HP fan	Standby
CHWP-1/2	Primary CHW Pump	2,800 GPM	80 ft TDH, 75 HP	VFD	Lead / Standby
CHWP-3/4	Secondary CHW Pump	3,500 GPM	60 ft TDH, 75 HP	VFD	Lead / Standby
CWP-1/2	Condenser Water Pump	4,200 GPM	60 ft TDH, 100 HP	VFD	Lead / Standby

3.0 SYSTEM SCHEMATIC / DRAWING



4.1 Chiller Performance

4.2 Cooling Tower

- Pressure test all piping at 1.5× working pressure prior to insulation.
- Insulate all CHW piping per ASHRAE 90.1 Table 6.8.3; vapor barrier continuous.
- Provide commissioning plan per ASHRAE Guideline 1.5 for chiller plant.
- Water treatment contract required prior to initial fill — submit 1 year proposal.
- Vibration isolation: spring-type for chillers and pumps, neoprene for CT fan motors.
- Coordinate cooling tower structural loading with structural engineer (roof or grade).